

Minjung Kim

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PROFESSIONAL SUMMARY

AI/ML Research Engineer with 4.5 years of production experience at Hyundai Motor Company deploying large-scale VLM inference systems, advancing LiDAR 3D perception, and building autonomous driving pipelines. Inventor on 10 patents and author of 2 publications in perception and domain adaptation. Expertise spans multi-GPU serving infrastructure (Ray Serve, H100 clusters, Tensor Parallelism), novel backbone design, and end-to-end auto-labeling systems at scale. M.S. in Automotive-Computer Convergence Engineering, GPA 4.0/4.0. Targeting a Big Tech AI lab research engineering role where VLM systems and perception R&D intersect.

EDUCATION

Hanyang University <i>M.S. in Automotive-Computer Convergence Engineering. Advisor: Professor Kunsoo Huh, Ph.D.. GPA: 4.0/4.0.</i>	Seoul, South Korea Feb. 2022
Hanyang University <i>B.S. in Mechanical Engineering. Cum Laude. GPA: 3.48/4.0.</i>	Ansan, South Korea Feb. 2020

EXPERIENCE

Data Engineer, Automotive Driving SW Development Team <i>Hyundai Motor Company</i> - VLM-Based Data Curation Research (2026) - Designed and deployed large-scale VLM inference servers using Ray Serve across 8 nodes of H100 GPUs, implementing Tensor Parallelism and Data Parallelism to serve Qwen3.5 9B and 122B models in production. - Improved server throughput from 430 to 5,000 scenes/hour by integrating Continuous Batching, Prefix Caching, and Chunked Prefill serving techniques. - Built a production-grade Traffic Light Detection pipeline on Ray Serve, benchmarking VLM, SAM3, and YOLO architectures to optimize signal recognition. - Scaled data curation and metadata verification workflows by adapting the SOTA T* algorithm to extract key-frames from scene directories, integrating it with an in-house Qwen VLM and YOLO pipeline. - Architected parallelized API handling for automated data curation workflows, enabling efficient processing of large-scale autonomous driving datasets. 3D Object Detection and Multi-Object Tracking Research (2024-2025) - Designed a dual-grid backbone combining short-range dense and long-range sparse feature maps to reduce domain gaps between near and far LiDAR points. - Integrated multi-frame inputs during network training to stabilize yaw estimation and improve robustness against occlusions. - Developed and automated large-scale 3D object detection auto-labeling pipelines with Dockerization, Jenkins CI/CD, Airflow scheduling, and Triton server deployment. - Diagnosed and resolved dataset and network issues including intensity-timestamp overwriting and low-confidence tail classes; improved detection performance through regression loss re-weighting, negative heatmap masking, and balanced long-range sampling. - Applied Gaussian Process Regression for probabilistic path prediction, reconnecting fragmented tracklets and reducing ID-switching during long occlusions beyond Kalman filter capabilities. - Developed a re-identification module using projected 3D bounding boxes and semantic segmentation, applying IoU-based occlusion analysis and key point/GMM similarity matching to prevent false associations. - Redesigned tracking structure from a fixed array to a dynamic list to address ID overflow and support scalable multi-object tracking. - Simulation Data Generation, Domain Adaptation, and Deployment (2023-2024) - Led development of a training simulation database in collaboration with Applied Intuition, generating 10,000+ images tailored to vehicle specifications and camera parameters. - Researched domain adaptation techniques focusing on active learning-based logging trigger logic and real-to-sim adaptation; leveraged generative AI to synthesize tail-class traffic light datasets for underrepresented classes. - Supported Level 4 pilot service deployments across RoboRide (Gangnam), RoboShuttle (National Assembly, Incheon Airport, Pangyo, Namyang R&D), and RoboDelivery (CJ Logistics).	Jan. 2022 – Present <i>Seoul, South Korea</i>
Graduate Research Assistant, Automotive-Computer Convergence Engineering <i>Hanyang University</i> - Derived 3D object position and class from Object Detection and Monodepth outputs, integrating results as camera sensor data to enhance cognitive performance. - Achieved real-time obstacle recognition on a 1/32 scale model car using the DBSCAN algorithm and YOLO.	Mar. 2020 – Feb. 2022 <i>Seoul, South Korea</i>

- Customized a pyramid-structured GAN in PyTorch for Super Resolution on LiDAR depth images; presented findings at the Korea Automotive Engineers Association.
- Developed a shared encoder-branched decoder architecture to enable simultaneous Object Detection, Semantic Segmentation, and Depth Estimation for a graduation thesis.
- Performed map-based object filtering, tracking, and risk assessment for the Korea Automobile Research Institute autonomous driving project, verified through PG testing.
- Designed a network to estimate vehicle slip angle using sensor data for the Hyundai Motor stuck-situation judgment task.
- Implemented a TCP/IP-based V2V interface for autonomous driving integration in the Korea Automotive Research Institute autonomous driving system project.

Intern, Color Filter Group

Jul. 2018 – Jul. 2018

LG Display Company

Paju, South Korea

- Executed a Six Sigma-based quality improvement project, optimizing nozzle start time to reduce contour variation by 28% in the Color Filter production process.

Undergraduate Research Assistant, Department of Mechanical Engineering

Mar. 2015 – Feb. 2020

Hanyang University

Seoul, South Korea

- Designed and developed an electric board with MPU6050 sensors, resolving data overshoot issues through Kalman and complementary filters; earned the Excellence Award at the College of Engineering Academic Conference.
- Designed a rotary hydraulic supply unit for the Machine Tool Design Competition, selecting standard components, specifying tolerances, and implementing the design in AutoCAD with 3D modeling.
- Analyzed the efficiency of 3-link Scotch-Yoke and 4-link can compressor mechanisms using theoretical calculations, CATIA modeling, and MATLAB simulations.
- Designed and manufactured 'Smooth Crutches' using elastic springs and curved surfaces to enhance user comfort, optimizing materials for cost-efficiency and customizing spring stiffness for varying user weights.

CERTIFICATIONS

- Udacity Deep Reinforcement Learning Nanodegree Program (Sep. 2021)
- Udacity Sensor Fusion Nanodegree Program (Dec. 2020)
- Microsoft Office Specialist Expert - Excel (Aug. 2018)
- Six Sigma Green Belt Certificate (May 2017)

AWARDS & HONORS

- Best Female Researcher Award, The Korean Society of Automotive Engineers (2021)
- Brain Korea 21 (BK21) FOUR Fellowship, National Research Foundation of Korea (2021)
- Brain Korea 21 (BK21) RA/TA Fellowship, National Research Foundation of Korea (2020)
- Academic Excellence Scholarship (Hanyang-in), Hanyang University, 4 times (2020-2021)
- Hanyang ERICA Engineering Excellence (HEEE) Dean's Award, Cum Laude, Hanyang University (2020)
- Scholarship for Dormitory Accommodation, Hanyang University (2020)
- Scholarship for Volunteer Activities, Samsung and Green Umbrella ChildFund (2019)
- Mechanical Conference Grand Award, Hanyang University (2018)
- College of Engineering Conference Excellence Award, Hanyang University (2018)
- Creative Korea (CK-II) Tutoring Scholarship, National Research Foundation of Korea, 2 times (2018)
- Scholarship for Internship Program, Hanyang University (2018)
- Mechanical Conference Grand Award, Hanyang University (2017)
- Applied Excellence Scholarship, Hanyang University (2015)
- National College Entrance Exam Merit Scholarship, Hanyang University (2015)

SKILLS

Vision-Language Models and Generative AI: VLM (Qwen3.5 9B/122B, large-scale inference), GANs, Generative AI for data synthesis, SAM, Semi-supervised Learning

MLOps and Serving Infrastructure: Ray Serve, Tensor Parallelism, Data Parallelism, Continuous Batching, Prefix Caching, Triton Inference Server, Docker, Jenkins CI/CD, Apache Airflow

Perception and Robotics: LiDAR 3D Object Detection, Multi-Object Tracking, Semantic Segmentation, Depth Estimation, Sensor Fusion, YOLO, DBSCAN, Kalman Filter, Gaussian Process Regression

Machine Learning and Deep Learning: PyTorch, Domain Adaptation, Active Learning, Object Detection

Programming and Tools: Python, MATLAB, AutoCAD, CATIA, TCP/IP

PUBLICATIONS

- Minjung Kim, Juheon Baeg and Hyunkyu Lim, "Bridging the Range Gap: Semi-Supervised Learning for Long-Range Detection," 33rd Hyundai Motor Company Conference, June 27, 2025.
- Minjung Kim, Seungwon Choi and Kunsoo Huh*, "LiDAR Super-resolution based on Multi-scale Generative Adversarial Network," KSAE Annual Spring Conference, Jeju, South Korea, June 23-26, 2021 (Awarded).

PATENTS

- Minjung Kim, "Domain-consistent Learning via Confidence-guided Pseudo-labels and Spatial Masking," South Korea, In process.
- Minjung Kim, "Distance-Balanced Sampling with Adaptive Tolerance Ray Casting for Lidar Networks," South Korea, In process.
- Minjung Kim, "Sensor-Specific Voxelization and Fusion in Multi-LiDAR Perception," South Korea, KR No. 1020260099136, 2026.
- Minjung Kim, "Keep your Road Elements Straight," South Korea, KR No. 1020250156424, 2025.
- Minjung Kim, "Bring your simulation to real world," South Korea, KR No. 1020250140279, 2025.
- Minjung Kim, Yohan Sung, "Occlusion detection by estimating the hidden space using 3D modeling," South Korea, KR No. 1020250142081, 2025.
- Minjung Kim, "Yaw stabilization method for offline tracking," South Korea, KR No. 1020250147907, 2025.
- Minjung Kim, Woocheol Choi, Hyunkyu Lim, "Tracklet re-connection method based on probabilistic model," South Korea, KR No. 1020250112467, 2025.
- Woocheol Choi, Minjung Kim, Hyunkyu Lim, "Tracking re-identification method for auto labeling," United States and South Korea, US No. 19/663,765, KR No. 1020250072080, 2025.
- Minjung Kim, Jaewon Key, Hyunkyu Lim, "Look back on your tracking," United States, China and South Korea, US No. 18/895,120, CN No. 202411470342.X, KR No. 1020230180170, 2023.

TEACHING

- Graduate Teaching Assistant, Automotive-SW Convergence Seminar 1, Hanyang University, Seoul, South Korea (Spring 2021): Answered student questions in person and online, marked assignments, and proctored exams.
- Tutor, Hanyang University, Ansan, South Korea: Solid Mechanics for sophomores (Spring 2019); Machining Process and Design for juniors (Fall 2018); Applied Fluid Mechanics Design for juniors (Spring 2018).

LEADERSHIP

- Executive Director, Department of Mechanical Engineering Students Council (30th College of Engineering Student Council), Hanyang University, Ansan, South Korea (Jan. 2017 - Dec. 2018): Organized college events including freshman orientation, workshops, and campus tours.
- Executive Director, Mechanical Engineering Academic Club RED, Hanyang University, Ansan, South Korea (Jan. 2017 - Dec. 2018): Served as a MATLAB study mentor and collaborated with peers in multiple academic competitions.